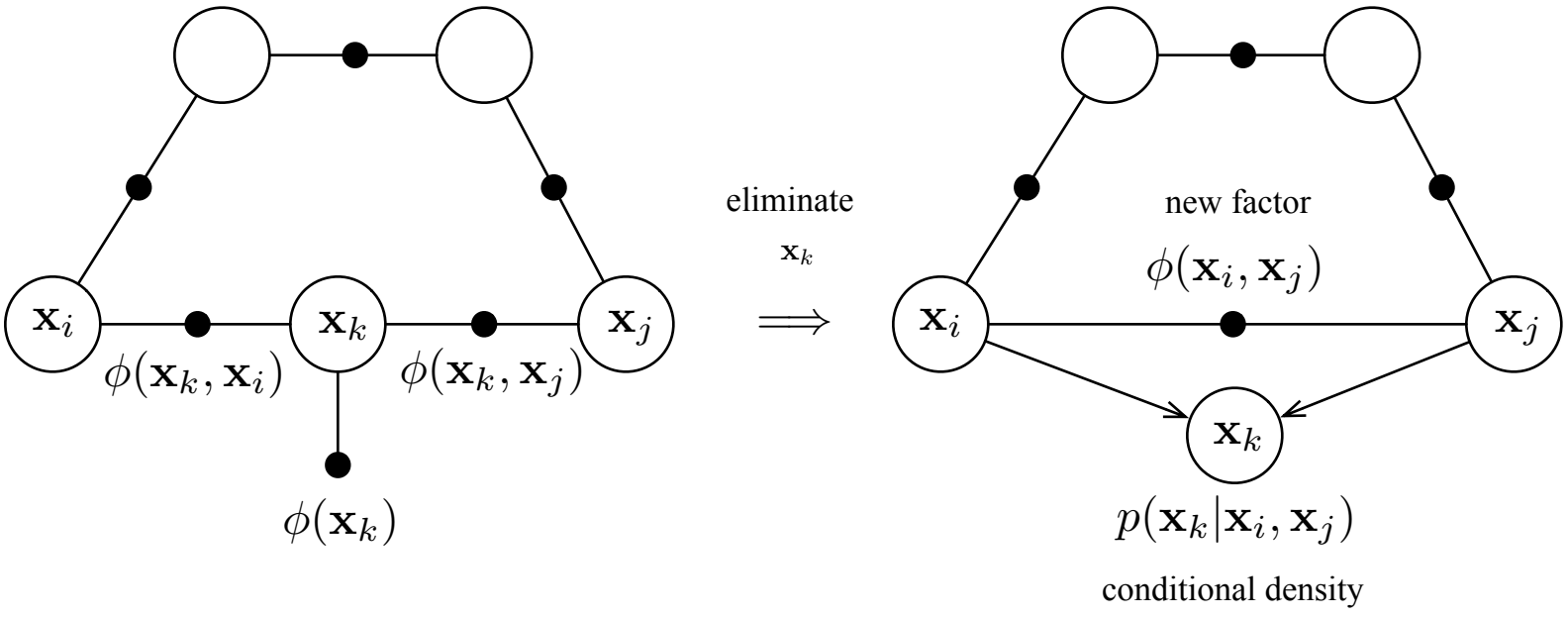
$\mathbf{x}_i$
$\mathbf{x}_k$
$\mathbf{x}_j$
$\phi(\mathbf{x}_k, \mathbf{x}_i)$
$\phi(\mathbf{x}_k, \mathbf{x}_j)$
$\phi(\mathbf{x}_k)$
eliminate
$\mathbf{x}_k$
$\Longrightarrow$
new factor
$\phi(\mathbf{x}_i, \mathbf{x}_j)$
$\mathbf{x}_i$
$\mathbf{x}_j$
$\mathbf{x}_k$
$p(\mathbf{x}_k | \mathbf{x}_i, \mathbf{x}_j)$
conditional density